

Diseases Of Airways + Bronchial Asthma

Asthma guideline map for patients >12 years: severity, control, treatment steps

MED401-2 Chest taken

Severity: Not On Controllers				
Domain	Intermittent	Mild persistent	Moderate persistent	Severe persistent
Symptoms	<2 days/week	>2 days/week not daily	Daily	Throughout day
Night awakening	<2x/month	3-4x/month	>1x/week not nightly	Often 7x/week
SABA use	<2 days/week	>2 days/week not >1x/day	Daily	Several times/day
Activity	None	Minor limitation	Some limitation	Extremely limited
FEV1	>80%, normal ratio	>80%, normal ratio	>60% and <80%; ratio down 5%	<60%; ratio down >5%

First case logic

- 19-year-old with EIB, perennial congestion, no daytime symptoms, night symptoms 2/month, albuterol pre-exercise only.
- Moderate exercise limitation despite albuterol makes severity moderate persistent.
- Recommended treatment in case: low-dose ICS plus LABA; also nasal steroid, SABA PRN, action plan, technique and avoidance education.

Asthma Control			
Domain	Well controlled	Not well controlled	Very poorly controlled
Symptoms	<2 days/week	>2 days/week	Throughout day
Night awakening	<2/month	1-3x/week	>4x/week
Activity	None	Some limitation	Extremely limited
SABA	<2 days/week	>2 days/week	Several times/day
FEV1 / peak flow	>80%	60-80%	<60%
Exacerbations	0-1/year	>2/year	>2/year

Rules of Two

- More than two quick-relief inhaler uses per week?
- Night asthma more than two times per month?
- Day symptoms more than twice a week?
- Attacks more than twice a year?
- Any limitation on exercise or quality of life?

Stepwise Management >12 Years		
Step	Preferred	Notes
1	SABA PRN	Intermittent asthma.
2	Low-dose ICS	Alternatives: cromolyn, LTRA, nedocromil, theophylline.
3	Low-dose ICS + LABA OR medium-dose ICS	Consider consultation.
4	Medium-dose ICS + LABA	Consult specialist if step 4 or higher.
5	High-dose ICS + LABA	Consider omalizumab for allergic patients.
6	High-dose ICS + LABA + oral steroid	Consider omalizumab.

Escalation principles and updates

- Before stepping up: check adherence, inhaler technique, and environmental control.
- Step down when asthma is well controlled for at least 3 months.
- SABA more than twice weekly for symptom control indicates inadequate control.
- Later developments include tiotropium add-on, ED ipratropium, mepolizumab for severe eosinophilic asthma, omalizumab, bronchial thermoplasty evidence, exercise benefit, vitamin D note, and limited macrolide benefit.

Lecture take-home

- Classify severity on the first visit, then assess control on follow-up visits.
- Assess both impairment and risk: symptoms, night awakening, activity, SABA use, lung function, and exacerbations.
- ICS are stressed as the drug of first choice for persistent asthma.
- For severe uncontrolled asthma, the lecture highlights omalizumab or mepolizumab; for moderate asthma it mentions zileuton.